

Chidera Onyebu

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EDUCATION

University of Maryland

Bachelor of Science in Computer Science and Applied Mathematics
CGPA: 3.7

College Park, Maryland

Aug. 2024 – May 2028

Jesuit Memorial College

High School Degree

Lagos, Nigeria

Sept. 2018 – May 2024

PROJECTS

Apex — Event-Driven Backtesting Framework | *Python*

June 2026 – Present

- Built an event-driven backtesting engine that structurally prevents lookahead bias (signals on bar t fill at bar $t+1$'s open) with commission, spread, and market-impact cost modeling; 51 tests, strict typing.
- Implemented a portfolio risk layer: volatility-targeted position sizing, exposure caps, and a max-drawdown kill switch that detects at close, flattens at next open, and logs every intervention as an auditable event.
- Added multiple-testing statistics (Probabilistic & Deflated Sharpe Ratio): after a 24-configuration parameter sweep, the framework correctly rejected its own best in-sample strategy (raw PSR 96% $\rightarrow$ 89.5% deflated) on data with no real edge.
- Walk-forward validation surfaced that a strategy beating buy-and-hold in-sample (+2.2 pts) lost out-of-sample in 4 of 6 folds — the honest result the framework exists to produce.

Nutrition5k Data-Leakage Audit & Calorie Classifier | *Python, PyTorch · AI4ALL*

June 2026 – Present

- Discovered session-level data leakage in a Google Research dataset split that standard sample-ID checks pass: 94.7% of test images shared a capture session (lighting, camera, table) with training images.
- Quantified the effect with a controlled 20-run experiment (5 seeds $\times$ 2 architectures, common held-out test): the leaky split inflated reported accuracy by +3.3 points, positive in 10/10 runs and architecture-independent.
- Trained an ordinal multi-task CNN on a clean session-grouped split to 74.2% test accuracy (2.2 $\times$ the 33.3% baseline), with the test set evaluated exactly once; authored a reproducible write-up of the finding.
- Shipped the team's Streamlit deployment (merged PR): photo upload $\rightarrow$ prediction with per-threshold probabilities and Grad-CAM saliency maps; 71 tests across the pipeline.

TradeDesk — LLM Paper-Trading Agent | *Python, LangGraph, FastAPI*

June 2026 – Present

- Built an intent-gated trading agent for the ServiceNow AI.Accelerate capstone, where LLM output is validated data, never an executable command: every trade requires explicit human confirmation in a separate turn, and roles are enforced in code, not prompts.
- Integrated Alpaca's paper-trading API with idempotent order submission — a retry after timeout returns the original fill rather than double-executing — and a hard-coded paper-only endpoint with no configuration path to live funds.
- Authored an 18-scenario adversarial evaluation suite (prompt injection, fake authority claims, role escalation, stale confirmations) asserting at the broker level: 0 unauthorized executions across all scenarios; 226 tests total.
- Exposed the agent as an MCP server with a two-step propose/confirm token handshake (single-use, 120-second expiry), preserving human-in-the-loop review across protocols.

EXPERIENCE

ColorStack - Member and Peer Mentor

University of Maryland

September 2024 – Present

College Park, Maryland

- Mentor 20+ CS students one-on-one through coursework and projects in a national community of 16,100+ members; run recurring study sessions and debugging workshops.

Alpha Lambda Delta - Member and Academic Mentor

University of Maryland

April 2025 – Present

College Park, Maryland

- Lead review sessions for 50+ first-year students in a national honor society admitting the top 20% of the class — tutoring, exam prep, and course-selection advising (8+ sessions to date).

TECHNICAL SKILLS

Languages: Java, Python, JavaScript (ES6+), HTML/CSS, C, Bash

ML & Data: PyTorch, torchvision, scikit-learn, NumPy, pandas, Matplotlib, Pillow, Grad-CAM, Streamlit

Backend & Agents: FastAPI, LangGraph, Anthropic Claude API (structured outputs), Model Context Protocol (MCP), Pydantic, httpx, Uvicorn, REST APIs (Alpaca)

Developer Tools: Git, GitHub (pull requests, Pages, releases), pytest, ruff, mypy, Kaggle (GPU kernels, datasets API), VS Code, Unix/zsh

Quantitative Methods: Event-driven backtesting, walk-forward validation, volatility targeting, probabilistic & deflated Sharpe ratio, cost/slippage modeling